ONSSET SCENARIO ANALYSIS · UGANDA · 2024 → 2030

Universal electricity access in Uganda: costs, technology mix and the effect of key assumptions

Six least-cost electrification scenarios on Uganda's national settlement dataset (397,910 settlements). They test the energy-cost assumption (±50%) and a partial 75% access target under three prioritisation rules.

Model: OnSSET 2.1 (commit 486c260) · Scenarios run 28 Oct 2026 · Demand: Tier 4 urban, Tier 3 rural, Tier 1 small rural (<100 HH)

$6.80 B
Base-case investment to reach 100% access by 2030

8.15 M
New household connections (from 35% access in 2024)

36%
Share of 2030 population served off-grid (mostly stand-alone PV)

−30%
Investment saved by a 75% target with road- or travel-time-based priority

Key findings

1. **The base case is a grid-plus-solar split.** At $0.35/kWh, 64% of Ugandans in 2030 are served by the grid: 23.3 M through densification of already-connected areas and 13.9 M through new MV extensions. Stand-alone PV covers 35% (20.6 M people) and mini-grids just 0.8%. Grid extension accounts for 59% of the $6.80 B.
2. **Higher energy costs have a much bigger effect than lower ones.** Cutting the cost by 50% changes almost nothing: investment falls 0.8% and about 0.26 M people move from mini-grids to the grid. Raising it by 50% moves about 1.85 M people from grid extension to PV-hybrid mini-grids, so mini-grid coverage grows fivefold (0.44 M → 2.28 M). Investment rises 4.9% to $7.13 B and grid emissions fall 6.5%. The baseline sits close to the tipping point where mini-grids beat grid extension, but well clear of the one in the other direction.
3. **Stand-alone PV doesn't respond to the energy cost.** Its 20.6 M users stay the same across ±50%. These are small, dispersed Tier 1 settlements where grid extension is never competitive, so the off-grid solar programme is the most robust part of the plan.
4. **A 75% target saves up to 30%, but the saving depends on the prioritisation rule.** Rules 1 (road distance) and 2 (travel time) cost about $4.7 B, while rule 3 (largest settlements first) costs $6.02 B, 27% more. Rule 3 reaches the same 75% through a full-scale grid build-out ($1,149 per new connection against about $900), so it delivers grid coverage and emissions close to the 100% case while 14.6 M people stay unconnected.
5. **The prioritisation rule decides who is left behind.** Rule 1 follows road corridors and spreads access widely, with 8 of 112 districts below 50%. Rule 2 concentrates on well-connected zones: 40 districts fall below 50%, and Kotido and Kaabong in Karamoja get nothing. Rule 3 is the most even geographically (no district below 27%), but leaves out about 364,000 small settlements, 92% of all settlements.

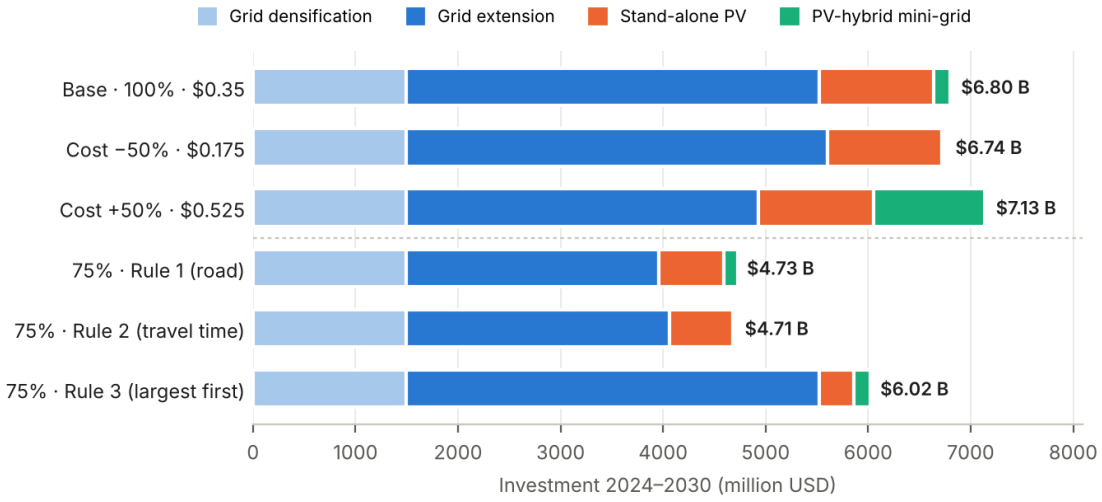
> **Bottom line.** The energy-cost assumption mainly decides how big a role mini-grids play, and only when costs rise. The access target and prioritisation rule drive both the budget and the equity of the outcome. For a partial target, road-based priority (rule 1) gives the lowest cost with the fewest districts left far behind.

SCENARIO COMPARISON

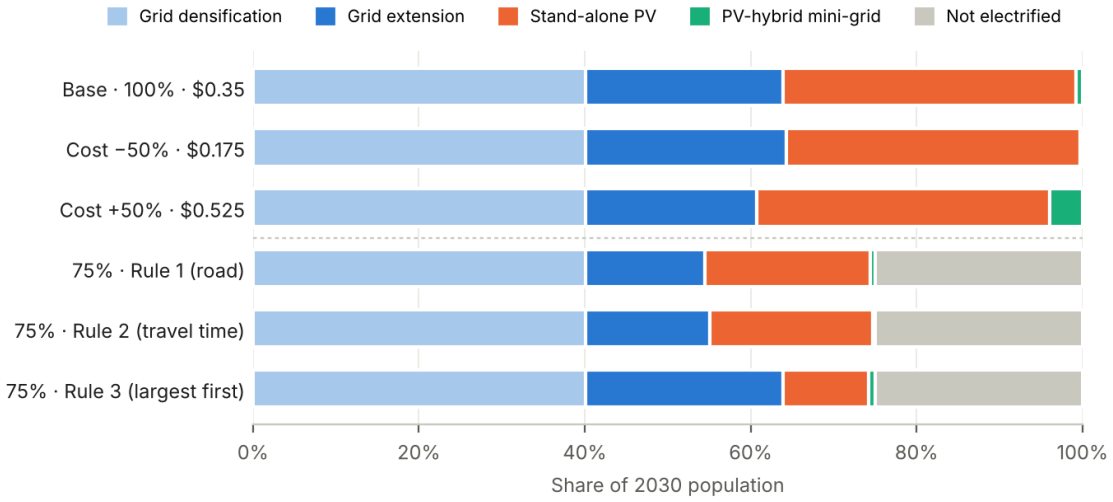
All six scenarios at a glance

Scenario	Access 2030	Investment (US$ B)	vs base	New conn. (M HH)	US$ per conn.	Grid (M people)	SA PV (M people)	Mini-grid (M people)	Unserved (M people)	Added cap. (MW)	Grid CO_2 (kt/yr)
Base · 100% · $0.35	100%	**6.80**	—	8.15	834	37.21	20.60	0.44	0	763	1,894
Cost −50% · $0.175	100%	6.74	−0.8%	8.15	827	37.47	20.60	0.18	0	746	1,916
Cost +50% · $0.525	100%	7.13	+4.9%	8.15	875	35.35	20.62	2.28	0	856	1,771
75% · Rule 1 (road)	75%	4.73	−30.4%	5.24	902	31.70	11.62	0.37	14.55	616	1,557
75% · Rule 2 (travel time)	75%	4.71	−30.7%	5.24	899	32.11	11.40	0.18	14.56	592	1,552
75% · Rule 3 (largest first)	75%	6.02	−11.5%	5.24	1,149	37.20	6.04	0.44	14.56	728	1,894

Investment and capacity are totals for 2024–2030. "Grid" covers densification (23.34 M people, the same in every scenario) plus extension. Emissions are annual emissions from grid generation in 2030 at 745 gCO_2/kWh; off-grid PV options are zero-emission in the model. US$ per connection = total investment ÷ new household connections.



Investment by technology. Densification ($1.49 B) is fixed; the scenarios differ in grid extension and off-grid spending.

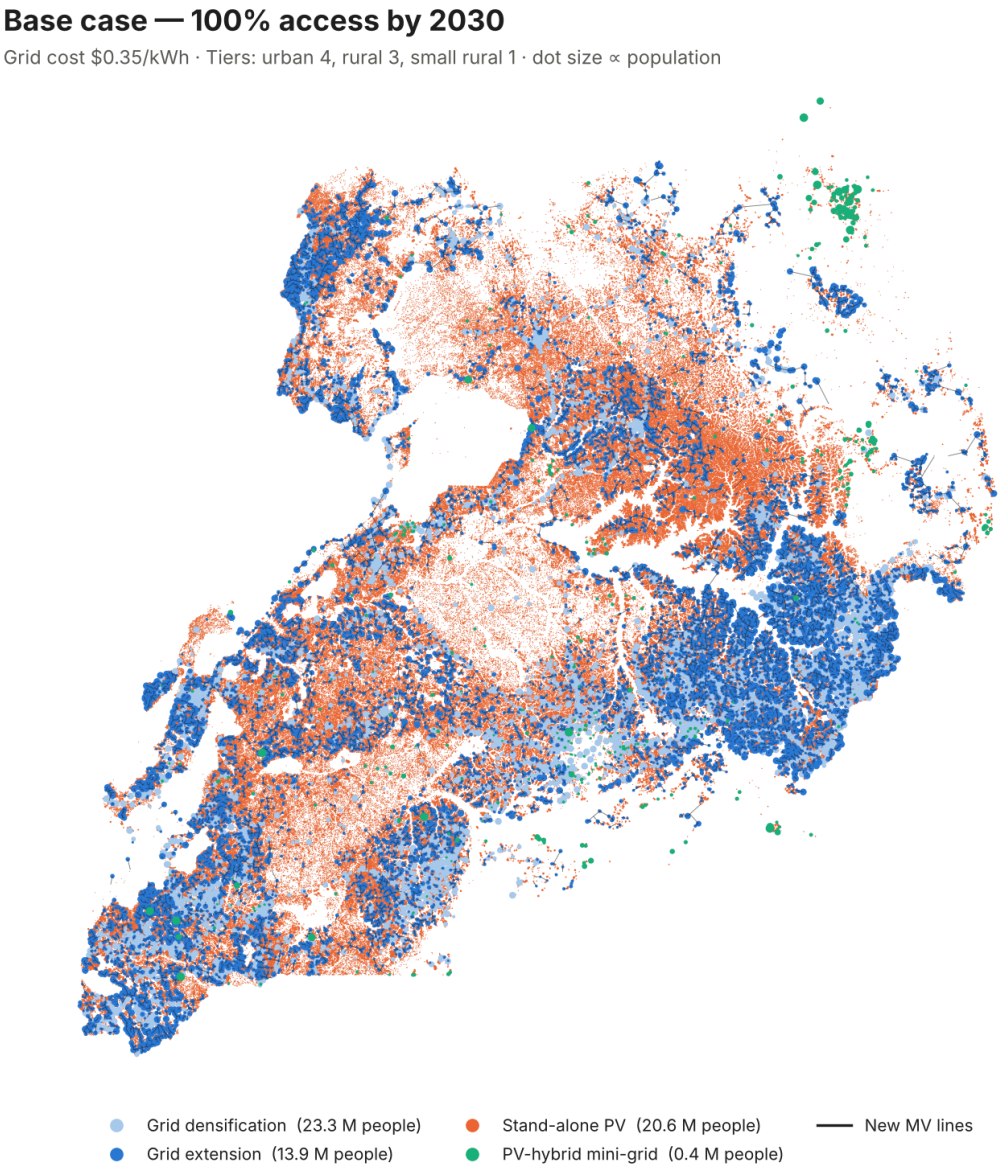


How the 2030 population (58.25 M) is served in each scenario.

SCENARIO 0 · BASE CASE

Base case: 100% access at $0.35/kWh

The least-cost plan extends the MV network through the densely populated south and east (the Kampala–Lake Victoria belt and the Mbale/Busoga region), the southwest highlands and the West Nile sub-region. Most of the northern and central rural interior, where settlements are small and spread out, is served by solar home systems. PV-hybrid mini-grids are cost-effective only in a few clusters, mainly in Karamoja in the northeast, where large settlements are far from existing MV lines.



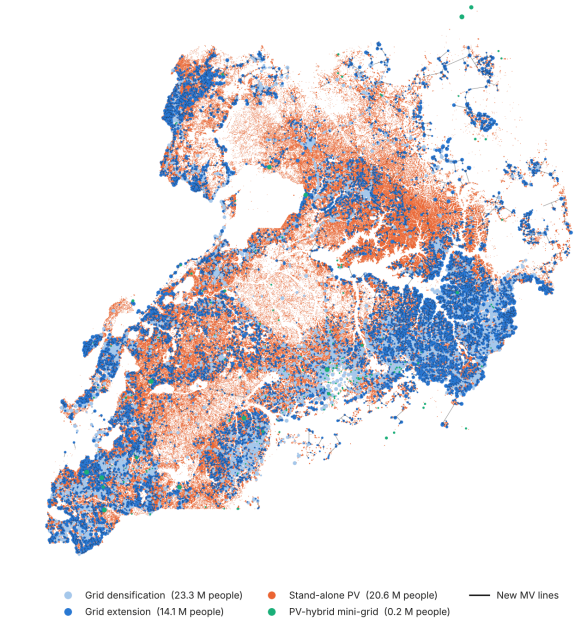
Least-cost technology by settlement in 2030 (dot size ∝ population) with new MV lines.

WHAT-IF 1 · ENERGY COST ±50%

Energy cost: mini-grids are the swing technology

Low energy cost — $0.175/kWh (−50%)

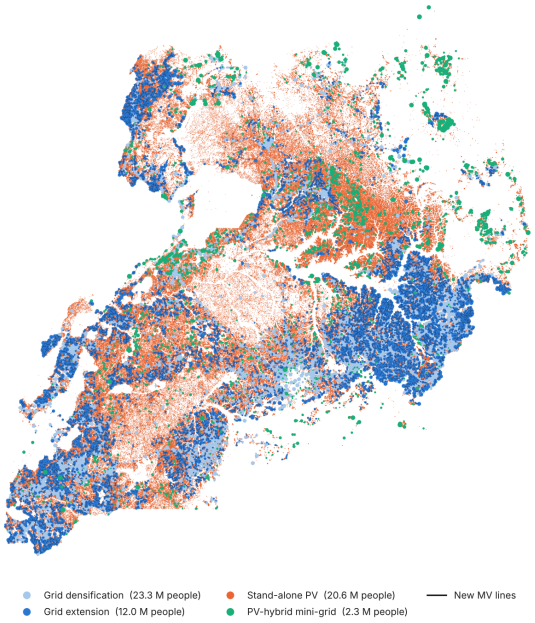
100% access by 2030 · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population



$0.175/kWh (−50%)

High energy cost — $0.525/kWh (+50%)

100% access by 2030 · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population



$0.525/kWh (+50%)

What changes

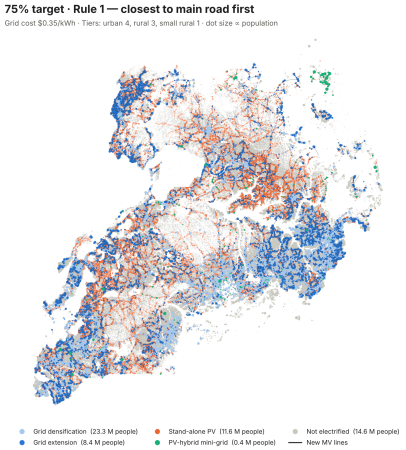
Metric	−50% ($0.175)	Base ($0.35)	+50% ($0.525)
Investment (US$ B)	6.74 (−0.8%)	6.80	7.13 (+4.9%)
Grid extension (M people)	14.13	13.87	12.02
PV-hybrid mini-grids (M people)	0.18	0.44	2.28
Stand-alone PV (M people)	20.60	20.60	20.62
Mini-grid investment (US$ M)	26	160	1,082
Grid CO_2 (kt/yr)	1,916	1,894	1,771

Interpretation

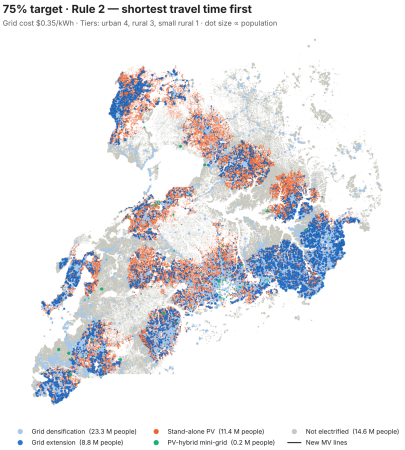
- **The response is lopsided.** Halving the cost shifts about 0.26 M people; raising it by half shifts about 1.85 M. Many medium-sized rural settlements in the north, Teso and Karamoja sit just below the grid-vs-mini-grid break-even at $0.35/kWh. A higher energy cost pushes them over it; a lower one has few marginal settlements left to win.
- **Densification and stand-alone PV are insensitive** over this range. Already-connected areas stay on the grid regardless, and Tier 1 households in small settlements are cheapest to serve with solar home systems at any of these prices.
- **Higher costs shift spending toward capital-heavy, zero-emission systems:** 93 MW more capacity in total, mostly PV and batteries, and 6.5% lower grid emissions.

WHAT-IF 2 · 75% ACCESS TARGET

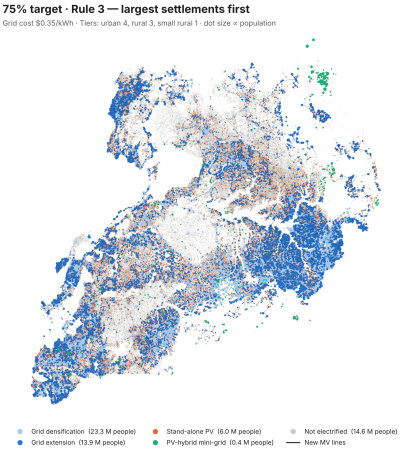
75% target: the prioritisation rule shapes cost and equity



Rule 1 · closest to main road



Rule 2 · shortest travel time



Rule 3 · largest settlements first



Spread of district-level access in 2030 under each rule (national rate is 75% in all three).

Rule	Investment (US$ B)	US$ per conn.	Grid ext. (M)	SA PV (M)	Settlements left unserved	Districts < 50%	Lowest district
1 · Road distance	4.73	902	8.37	11.62	182,900 (46%)	8	Buvuma 0% (island)
2 · Travel time	4.71	899	8.77	11.40	192,600 (48%)	40	Kotido, Kaabong 0%
3 · Largest first	6.02	1,149	13.87	6.04	364,200 (92%)	14	Kaberamaido, Otuke 27%

- **Rule 1** builds access along road corridors across all regions. Newly connected settlements sit on average 0.5 km from a main road, against 3.6 km for those left out. This gives the most geographically spread outcome at the lowest cost.
- **Rule 2** picks contiguous zones within reach of towns (average travel time 0.5 h, against 1.5 h for the unserved), so whole remote districts, especially in Karamoja, are left out. Its cost matches rule 1, but its equity is the worst of the three.
- **Rule 3** connects every large settlement, often with grid extension, and leaves out about 364,000 small hamlets (average 40 people each). It reproduces the 100% grid build-out ($4.03 B in extension), so it costs $1.3 B more than rules 1 and 2. In effect it front-loads the grid investment of universal access, and the remaining 25% could later be reached cheaply with stand-alone PV.

METHOD AND ASSUMPTIONS

Setup, assumptions and caveats

Data

- Calibrated national settlement file for Uganda (397,910 settlements; 49.9 M people and 35.0% access in 2024; 27.4% urban), with hourly PV and wind profiles from renewables.ninja and the existing MV network (3,216 line segments).

Scenario parameters

Parameter	Value	Note
Horizon	2024 → 2030	Start year set to the calibration year of the dataset; 2030 matches the SDG 7 target year
2030 population / urban share	58.25 M / 31%	Assumed (about 2.6%/yr growth); not specified in the brief
Demand tiers	Urban 4 · rural 3 · small rural 1	1,241 / 365 / 20 kWh per household per year; "small" = fewer than 100 households
Energy cost	$0.35/kWh (±50%)	OnSSET grid generation cost, i.e. the cost of supplying grid electricity, not a retail tariff
Access target	100%, or 75% with priority rule 1/2/3	Single step to 2030; no intermediate targets
Other inputs	OnSSET defaults	12% discount rate, 15% grid losses, PV $1,400/kW, battery $550/kWh, diesel $1.43/l, 745 gCO_2/kWh grid

Caveats

- The 2030 population and urban share are working assumptions. Changing them scales the results but is unlikely to change the conclusions about relative differences between scenarios.
- "Energy cost" is modelled as OnSSET's grid generation cost. Off-grid technology costs are unchanged across the ±50% runs, so the sensitivity captures the grid-vs-off-grid trade-off only.
- Investment covers new generation, network and off-grid capital needs for 2024–2030. It excludes operating costs and is not an LCOE comparison.
- Population totals in the 75% runs differ from 58.25 M by under 0.02% because of rounding in the engine's summary.
- Maps plot settlement centroids, not real network routes; new MV lines are the model's indicative layout.

Run log

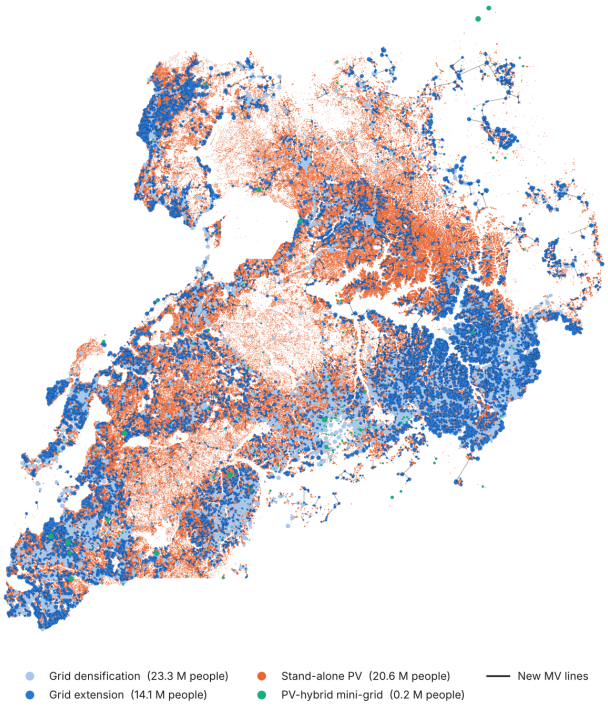
Run ID	Changed parameters	Runtime
uga_base_100_c035	—	2 m 39 s
uga_cost_low_c0175	grid_generation_cost = 0.175	2 m 16 s
uga_cost_high_c0525	grid_generation_cost = 0.525	2 m 39 s
uga_t75_prio1	end_electrification_rate_target = 0.75, prio_choice = 1	2 m 32 s
uga_t75_prio2	end_electrification_rate_target = 0.75, prio_choice = 2	2 m 35 s
uga_t75_prio3	end_electrification_rate_target = 0.75, prio_choice = 3	2 m 15 s

Full outputs (settlement results CSVs, new MV line GeoJSONs, parameter files) are in C:\Users\corra\Desktop\MCP\PowerMCP\OnSSET\runs\.

APPENDIX · FULL-SIZE GIS SOLUTIONS

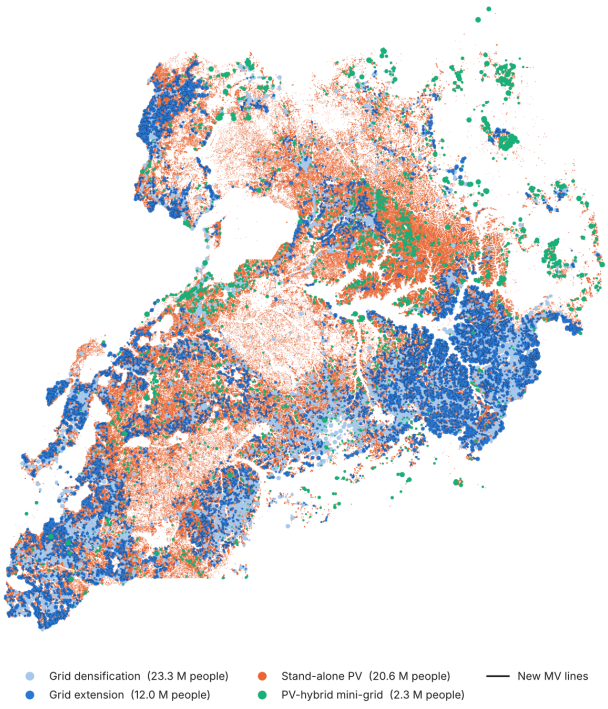
Low energy cost — $0.175/kWh (−50%)

100% access by 2030 · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population



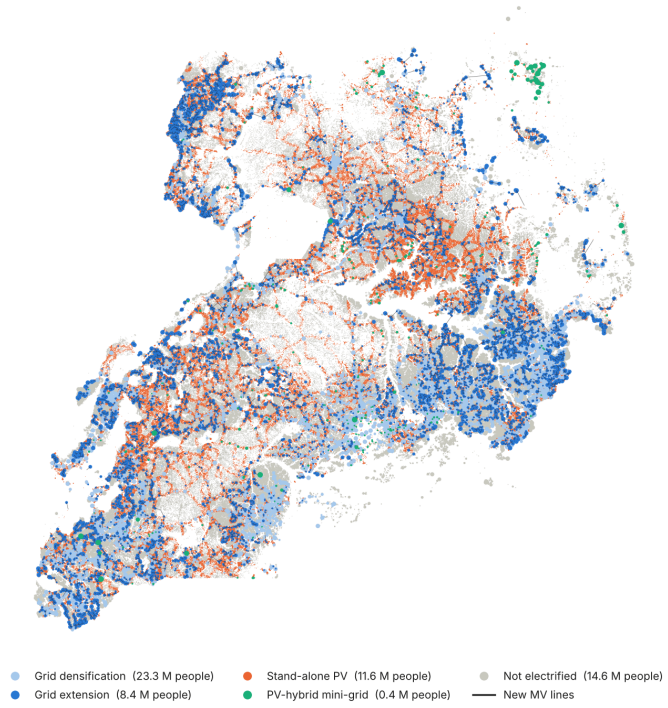
High energy cost — $0.525/kWh (+50%)

100% access by 2030 · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population



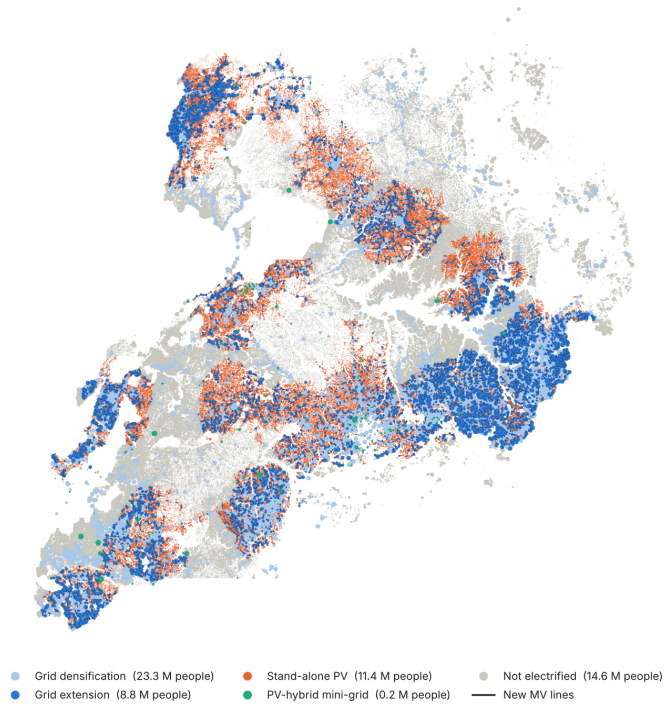
75% target · Rule 1 — closest to main road first

Grid cost $0.35/kWh · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population



75% target · Rule 2 — shortest travel time first

Grid cost $0.35/kWh · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population



75% target · Rule 3 — largest settlements first

Grid cost $0.35/kWh · Tiers: urban 4, rural 3, small rural 1 · dot size ∝ population

